

- 2 A light elastic string has natural length a and modulus of elasticity $4mg$. One end of the string is fixed to a point O on a smooth horizontal surface. A particle P of mass m is attached to the other end of the string. The particle P is projected along the surface in the direction OP . When the length of the string is $\frac{5}{4}a$, the speed of P is v . When the length of the string is $\frac{3}{2}a$, the speed of P is $\frac{1}{2}v$.

(a) Find an expression for v in terms of a and g . [4]

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(b) Find, in terms of g , the acceleration of P when the stretched length of the string is $\frac{3}{2}a$. [2]

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